The Door as an Interface for Immersion and Storytelling in Virtual Reality

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ABSTRACT

This paper presents a Virtual Reality (VR) experience that uses a physical door as an interface to deepen immersion and narrative involvement. By integrating a real door equipped with a tracker into a virtual environment, we create a tangible portal between four distinct virtual worlds. The project leverages the door's inherent affordances (opening, closing, passing through) to facilitate intuitive interaction while exploring its symbolic role as a threshold in storytelling. Developed in Unreal Engine 5 with HTC Vive hardware, the system employs mask-based rendering for seamless transitions. User tests confirm that the physical door anchors users in virtuality, enhancing spatial presence and emotional immersion. The findings indicate that physical interfaces can serve as effective anchors between real and virtual space and form an immersive framework for VR experiences.

INTRODUCTION

Bridging Realities: Physical Objects as Immersive Anchors in VR

Virtual Reality often disrupts presence through abstract interactions – controllers, menus, and gestures that lack tactile correspondence. This project addresses VR's "embodiment gap" by leveraging physical objects to deepen immersion. Inspired by MacLeod and Nordahl's "Two Rocks Do Not Make a Duck" (2019)[1], where tracked stones controlled environmental parameters in VR, we explore how mundane items can ground users in hybrid realities. Their work demonstrated that physical interfaces transform digital interactions from cognitive tasks into intuitive rituals. Another example are the "InflatableBots"[2], which enhance realism in VR reality by dynamically changing their position and size based on the happening inside the VR world. This approach creates real time haptic rendering for body scale objects.

Why a door?

As a universal cultural artifact, the door carries embodied knowledge: we understand its weight, sound, and resistance without instruction. Its mechanics – turning handles, pushing/pulling, crossing thresholds – are ingrained in daily life. This project repurposes that familiarity as an immersive affordance. When users touch a real door handle that aligns perfectly with its virtual counterpart, visuo-tactile congruence[3] creates moments of "meta-immersion": The users know that what they see is not real and yet they feel it. They recognize the staging and yet become part of it. The result is a moment of "meta-immersion": a deliberate break that is not disruptive but rather enables a deeper connection to the virtual experience.

This paper's contribution is a framework for designing object-driven VR immersion, validated through a door-based installation. We demonstrate how culturally embedded items can:

1. Reduce interaction barriers via inherent affordances
2. Amplify spatial presence through sensory alignment
3. Serve as modular interfaces for narrative environments

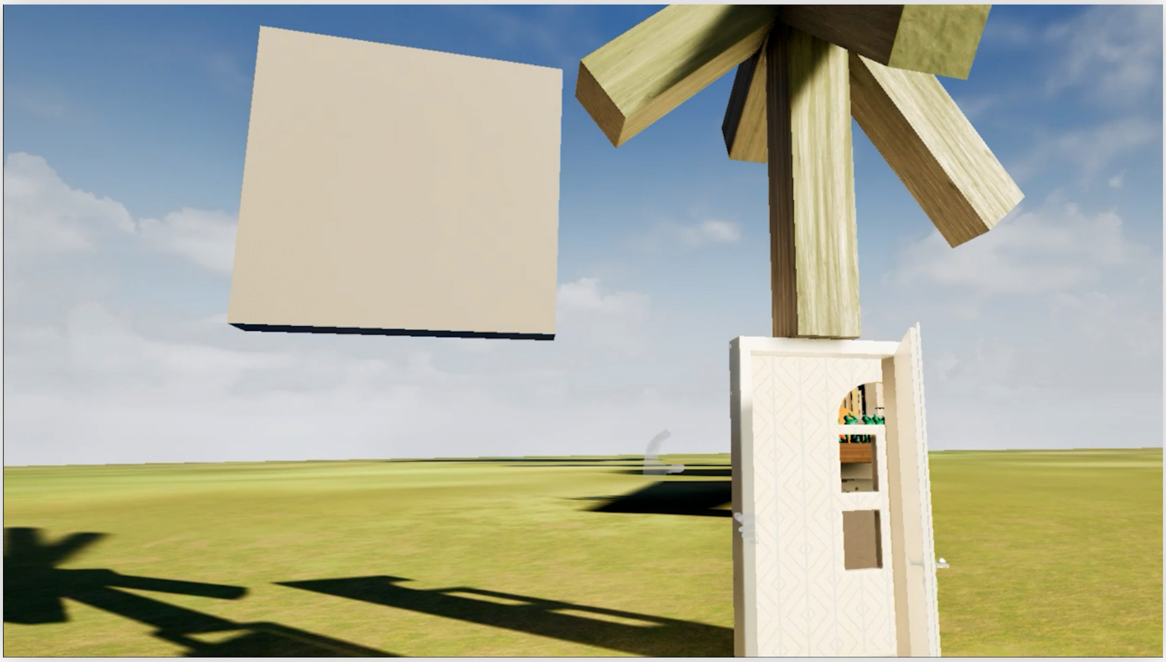
LIMINALITY AND THE THRESHOLD EXPERIENCE

The design of this VR installation is conceptually rooted in the anthropological notion of liminality – a transitional state of "in-between-ness." Originally introduced by Arnold van Gennep in his theory of Rites of Passage, the liminal phase marks the moment between separation and reintegration within societal transformation processes.[4] Victor Turner later expanded this framework, describing the liminal persona as a state in which individuals are no longer who they were, but not yet who they will become – a moment of ambiguity, vulnerability, and potential.[5]

This theoretical lens provides a symbolic foundation for using the door as a transitional interface in VR. In the context of spatial computing, liminality becomes more than a symbolic concept – it becomes an experiential condition. As users cross through the physical door and enter new digital environments, they undergo a moment of transformation that parallels the ritualistic crossing of thresholds described by Turner. Their spatial context, sensory input, and narrative positioning shift – triggering a new cognitive and emotional orientation.

Moreover, psychological studies like the Doorway Effect[6] demonstrate how passing through physical thresholds can disrupt short-term memory and cognitive continuity, reinforcing the notion that spatial transitions provoke mental resets. In this project, such transitions are not incidental but deliberately staged as narrative mechanisms. The door serves not only as a tool for locomotion but as a dramaturgical structure that activates liminal experience – a blend of spatial rupture, narrative realignment, and heightened awareness.

Through this integration of theory and practice, the door becomes both a literal and metaphorical threshold: it divides and connects, conceals and reveals, resets and transforms.



THE SYMBOLIC DIMENSION OF THE DOOR

Beyond its mechanical function, the door carries deep symbolic weight across cultures, media, and narrative forms. It operates as a threshold between states – inside and outside, known and unknown, self and other. In this project, the symbolic resonance of the door is central to its use as a narrative interface.

Drawing from Gaston Bachelard's Poetics of Space (1958), the door is viewed as a dialectical object – both opening and closing, inviting and forbidding. Bachelard writes: "The door schematizes two strong possibilities [...] It is a whole cosmos of the half-open."[7] The symbolic door thus activates imagination, desire, fear, and transformation – qualities especially potent in immersive environments.

This symbolism is reflected in narrative traditions across media. In Steppenwolf (Hesse, 1927), the door to the "Magic Theater" represents an inward journey into the fragmented psyche. In Alice in Wonderland (Disney, 1951), a small, ornate door signals the entry into an alternate reality – an invitation to the fantastical. In the video game Psychonauts (2005), doors are literal entry points into the minds of other characters, merging internal and external worlds. Together, these examples form a typology:

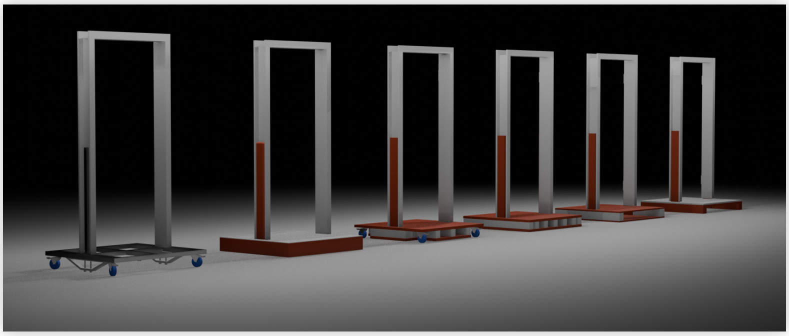
1. The door inward, toward introspection and memory
2. The door outward, toward adventure and the unknown
3. The door as mechanism, enabling mental or spatial transitions

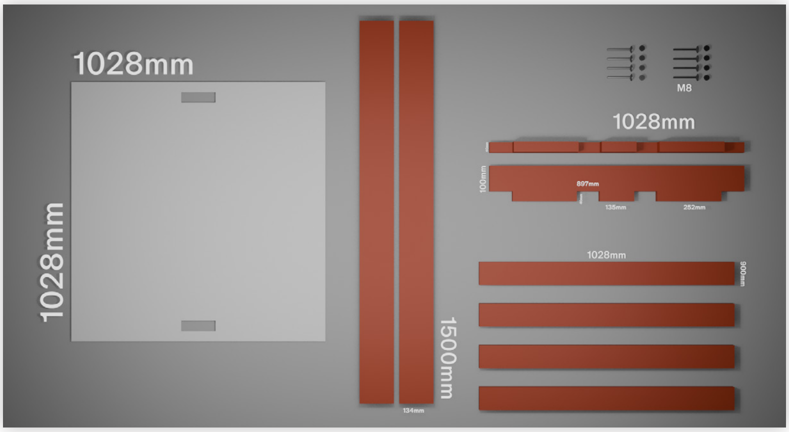
In this VR project, the physical door draws from this typology by linking symbolic transition with interactive mechanics. It transforms from a simple object into a dramaturgical hinge – shaping meaning not only through what it connects, but how it is crossed.

CONSTRUCTING THE PHYSICAL DOOR

The final version of the physical door interface was designed with modularity, stability, and precision in mind. Departing from the earlier prototype–built using a Euro pallet and metal pipes–the final setup was based on accurate measurements of the door frame and produced using high-quality materials and professional woodworking tools.

The base platform was constructed from recycled multiplex wood, with white-coated MDF panels forming the surface and sides. All elements were previsualized in 3D software and manufactured using CNC milling machines at the university workshop. The components were connected using only metric screws in combination with claw and insert nuts, enabling fast assembly and disassembly for transport and repeated use.





A hollow-core door was mounted within a wooden frame held by two upright support beams. This lighter alternative resolved stability issues caused by the heavy door leaf in the earlier prototype. Initially, a Vive Tracker was mounted to the upper outer corner of the door, where it remained visible to the Lighthouse tracking system while being visually unobtrusive. However, during user testing, it became apparent that the static behavior of the virtual door handle disrupted the sense of immersion.

To address this, a custom 3D-printed mount was designed to attach the tracker directly to the door handle. This approach allowed the tracker to capture both the rotation of the handle and the swinging movement of the door leaf. By excluding certain rotational axes in the VR engine logic, this setup maintained correct door behavior. Although the tracker became more visually prominent, the interaction became more consistent and believable.

TECHNICAL IMPLEMENTATIONS

Hardware

The hardware setup evolved significantly throughout the project. Initial prototypes employed a tethered HTC Vive Pro 2 headset and Vive Tracker 3.0, both reliant on an external Lighthouse tracking system. While this setup provided high precision, it limited user mobility: cables frequently interfered with the door frame, and the necessity of multiple tripods added logistical complexity.

Wireless alternatives were tested, including the Meta Quest 3, which offered high-resolution and sharp lenses. However, unstable incompatibility with external trackers led to performance issues.

The breakthrough came with the HTC Vive XR Elite in combination with the newly released Vive Ultimate Trackers. This setup provided native compatibility, wireless freedom, and robust inside-out tracking without the need for Lighthouses. The trackers used onboard cameras to map the physical environment and could be integrated directly into the VR pipeline. A dedicated Wi-Fi 6 router enabled a local network connection between headset and PC, ensuring high-speed, low-latency streaming.

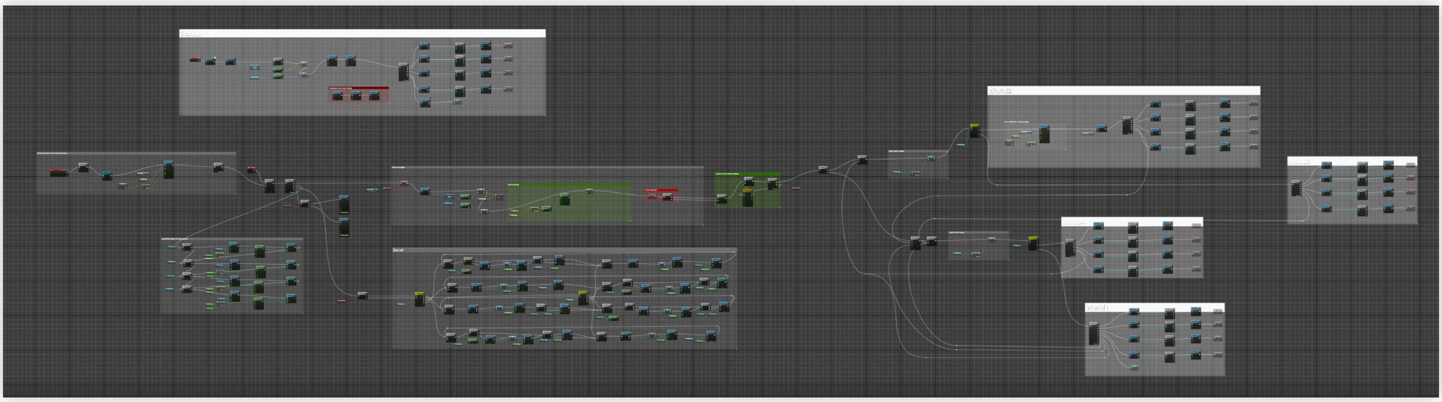
HTC Vive XR Elite

Software

The experience was developed using Unreal Engine 5, chosen for its high visual fidelity, flexible Blueprint scripting system, and prior user experience. A major technical challenge was creating a seamless portal system for transitioning between virtual worlds without loading screens or perceptual breaks.

A mask-based rendering technique was implemented.[8] Both scenes are loaded simultaneously and rendered conditionally using custom shaders and scene masks. This approach enabled convincing visual transitions and maintained immersion during spatial shifts.

Although initial plans included dynamic, non-linear world switching based on door interactions, technical limitations and unpredictable user movement led to the adoption of a linear progression model. Each passage through the physical door now triggers a controlled transition to the next virtual environment, guiding users through four interconnected worlds before returning to the original space. The door also acts as a trigger for ambient sound modulation: the angle of opening dynamically adjusts the volume and playback speed of the background music, while passing through the threshold initiates a seamless crossfade between the soundscapes of different worlds.



Level Blueprint Unreal engine 5.5

REFLECTIONS ON IMMERSION AND THE ROLE OF THE DOOR

Reflections on Immersion and the Role of the Door, one of the core insights gained through this project is that immersion in VR is not merely a technical phenomenon, but a dramaturgical and spatial one. Immersion is best understood not as a trick, but as an invitation–a framework that invites users to engage emotionally and physically at their own pace. The physical door, as an anchor point, offers more than interaction–it provides orientation, security, and continuity within the experience. Especially in abstract or unsettling scenes, its material presence helped users stay grounded.

During testing, both young and older participants reported a strong sense of immersion, despite differences in prior VR experience. Many expressed surprise at how "real" the environment felt once they passed through the door. The door acted not only as a literal threshold but as a narrative and emotional one.

Ultimately, the project demonstrates how a simple, familiar object can foster presence and narrative involvement in powerful and accessible ways.

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